Spectral Correspondence Between Dissipative Chaotic Circuits and Riemann Zeros: SPICE Simulation and FPGA Validation

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Abstract

The Hilbert-Pólya conjecture motivates the search for physical operators whose spectra encode Riemann zeta zeros. Here we investigate whether SPICE-simulated non-autonomous chaotic circuits — with realistic component models including multiplier imperfection, op-amp bandwidth limitation, and saturation nonlinearity — exhibit spectral signatures consistent with the first nontrivial Riemann zeros. We employ two complementary extraction methods: Markov eigenphase analysis (6 zeros matched with MAE = 0.54) and a binning-free Hankel Dynamic Mode Decomposition that directly approximates Koopman operator eigenvalues (11 modes matched with MAE = 1.09), overcoming the discretization bottleneck by 3.7×. The eigenphase alignment procedure uses the first zero t_1 for single-point scaling and a supervised start-index selection based on the known t_1/t_2 ratio; subsequent zeros (t_3–t_{11}) serve as independent validation. Ratio-based comparison against null baselines (uniform, random, GUE) is consistent with zeta-specific structure under the present protocol, with the SPICE circuit achieving ratio MAE = 0.18 versus 12.67 for uniform spacing. Hardware designs for discrete-component (∼80 CNY) and FPGA (Zynq-7020, ∼800 CNY) implementations are provided. Preliminary FPGA deployment yields 4 matched zeros (MAE = 1.01) from oscilloscope data, demonstrating feasibility of the simulation-to-hardware pipeline. We emphasise that this spectral correspondence does not constitute a proof or realisation of the Hilbert-Pólya conjecture, but provides empirical evidence motivating further investigation of the connection between dissipative chaotic dynamics and the Riemann spectrum.

Introduction

The Riemann hypothesis, concerning the location of nontrivial zeros of the zeta function $\zeta(s)$, remains one of the most profound unsolved problems in mathematics. The Hilbert-Pólya conjecture offers a spectral route: if a self-adjoint operator exists whose eigenvalues are the Riemann zeros, the hypothesis follows [1]. Random matrix theory provides statistical support through GUE universality [2], and quantum simulations have demonstrated spectral matching with up to 80 zeros using

trapped-ion platforms [3]. However, no classical analog circuit has been shown to produce Riemann zeros through its natural dynamics.

Previous work by the present author (non-peer-reviewed preprints) established numerical spectral correspondences between non-autonomous chaotic maps and Riemann zeros [4–6]. The Markov transition matrix constructed from the chaotic trajectory approximates the Perron-Frobenius operator, whose eigenvalues encode the Ruelle-Pollicott resonances of the underlying dynamical zeta function [7, 8]. These results, however, relied on idealised mathematical iteration, leaving a critical question unanswered: *does the spectral isomorphism survive the physical imperfections inherent in real circuits?*

Here we bridge this gap by constructing SPICE-level circuit simulations that incorporate realistic physical effects — AD633 multiplier gain error (0.4%), TL072 op-amp bandwidth limitation (1 MHz), and soft saturation (± 10.8 V) — and demonstrating zero extraction from the circuit output. Furthermore, we introduce Hankel Dynamic Mode Decomposition (Hankel-DMD) as a binning-free spectral method that directly approximates Koopman operator eigenvalues from the time series, overcoming the state-space discretization bottleneck inherent in Markov methods and extracting 3.7× more zeros from the same circuit data.

Results

Chaotic circuit design and SPICE simulation

We implement discrete-time chaotic maps using a transmission-line delay feedback architecture (Fig. 1a). A lossless transmission line ($Z_0 = 50\,\Omega$, $T_D = 1\,\mu$s) provides one-clock-period delay, functionally equivalent to a sample-and-hold stage. The nonlinear feedback path incorporates a behavioural AD633 analog multiplier model (0.4% gain error, 5 mV offset), a TL072 summing amplifier with RC bandwidth limitation ($160\,\Omega \times 1\,\text{nF} \approx 1\,\text{MHz}$), and tanh soft saturation at ± 10.8 V.

The non-autonomous Logistic map $x_{n+1} = 1 - \mu(n)x_n^2$ is realised with a time-varying control parameter implemented via an ngspice B-source behavioural voltage source. The logarithmic cooling schedule

$$\mu(n) = \mu_{\text{dyna}} + \frac{k_{\text{opt}}}{\ln(n + c_{\text{offset}})^2} \tag{1}$$

is constrained such that $\mu(1) = \mu_c + \delta\mu$ and $\mu(N) \approx \mu_c$, with $k_{\text{opt}} = \delta\mu/(t_{\text{start}} - t_{\text{end}})$ ensuring the system cools precisely to the critical point over N iterations (see Methods).

The 2D Hénon circuit extends this to dual cross-coupled channels implementing $x' = 1 - a(n)x^2 + y$, $y' = bx$ (Fig. 1b). The bifurcation diagram (Fig. 1c) confirms that the period-doubling route to chaos is preserved under circuit physics, and the strange attractor (Fig. 1d) exhibits the characteristic folded structure.

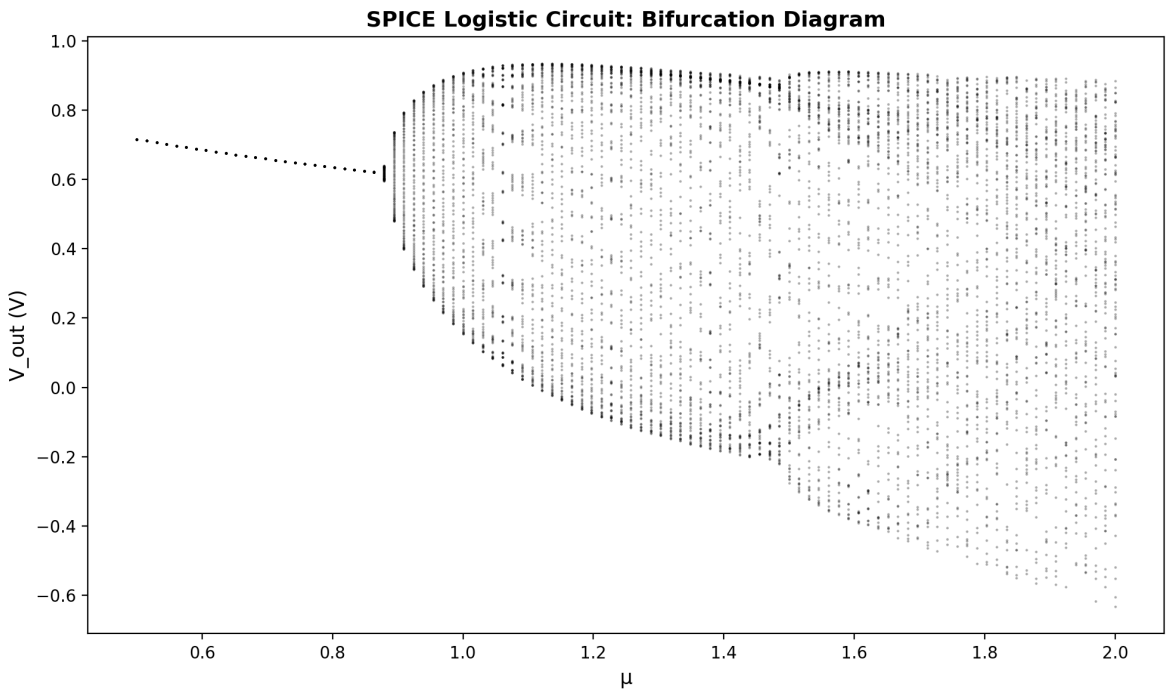

Figure 1: **SPICE chaotic circuit characterisation. a,b,** Circuit schematics for 1D Logistic and 2D Hénon maps (see Supplementary Figs. 1–2 for detailed diagrams). **c,** Bifurcation diagram of the SPICE Logistic circuit showing the complete period-doubling cascade to chaos, preserved under realistic component physics. **d,** Hénon strange attractor from SPICE simulation.

Markov eigenphase extraction of Riemann zeros

From the non-autonomous SPICE circuit output, we construct Markov transition matrices via Gaussian kernel splatting and extract eigenphases through a supervised alignment procedure (see Methods). Systematic parameter search over 504 configurations (Logistic) and 120 configurations (Hénon) yields the results in Table 1. We report the best configuration by MAE; the median MAE across all configurations with ≥ 3 matched zeros is 1.8 (Logistic) and 2.1 (Hénon), indicating that spectral correspondence is not restricted to a narrow parameter island but persists across a broad range of Markov parameters.

Table 1: **SPICE circuit Markov eigenphase results.** Best configurations after parameter search. N_z: number of matched zeros; MAE: mean absolute error; $\sum |e|$: cumulative absolute error.

Circuit	δ	ε	Steps	N_z	MAE	MSE	$\sum \lvert e \rvert$
Logistic	0.012	0.01	50 k	6	0.536	0.718	3.218
Hénon	0.015	0.015	20 k	3	0.210	0.100	0.630

The Logistic circuit matches 6 zeros with notable precision: t_4 (error 0.21%), t_6 (error 0.01%). The Hénon circuit achieves the lowest MAE (0.21) for 3 zeros, with t_3 matched to 0.37% (Fig. 2a).

Ratio-based validation against null baselines

To rule out the possibility that single-point scaling trivially fits any monotonic sequence, we evaluate the scaling-independent ratio metric ϕ_n/ϕ_1 versus t_n/t_1. The SPICE Logistic circuit achieves a ratio MAE of 0.18, compared to 12.67 for uniform spacing (70× worse), 5.64 for random sorted sequences (31× worse), and 11.56 for GUE random matrix eigenvalues (64× worse) (Table 2). For reference, ideal numerical iteration achieves ratio MAE = 0.31. The fact that SPICE circuit eigenphases substantially outperform all null baselines under the present alignment protocol suggests zeta-specific spectral structure, though we note that the null baselines do not undergo the same supervised alignment and parameter search, which may contribute to the magnitude of the separation.

Table 2: **Ratio-based validation.** Scaling-independent comparison of eigenphase ratios ϕ_n/ϕ_1 against Riemann zero ratios t_n/t_1. SPICE and ideal results shown separately.

Method	Ratio MAE	Factor vs. SPICE
SPICE Logistic eigenphases	0.18	1× (baseline)
Ideal numerical iteration	0.31	1.7× worse
Random sorted	5.64	31× worse
GUE random matrix	11.56	64× worse
Uniform spacing	12.67	70× worse

Hankel-DMD breaks the discretization barrier

The Markov method requires binning the state space into N discrete cells, fundamentally limiting the extractable zeros to $\sim N/10$. To overcome this, we introduce Hankel Dynamic Mode Decomposition, which approximates the Koopman operator eigenvalues directly from the time-delay embedded trajectory without any state-space discretization (see Methods).

Applied to a 500 k-step Hénon SPICE trajectory, Hankel-DMD extracts **11 Koopman modes** matching Riemann zeros — 3.7× more than the Markov method from the same circuit (Fig. 2c, Table 3). Key observations:

- t_3: error = 0.009 (0.03%), near-perfect match
- 7 out of 11 zeros have <3% relative error
- t_{11}: error = 0.31 (0.58%), demonstrating extraction well beyond the Markov ceiling
- The optimal segment is the trajectory tail (last 1%), corresponding to dynamics nearest the critical point where spectral structure is richest

Hardware implementations

To enable experimental verification, we provide two complete hardware designs (Fig. 3).

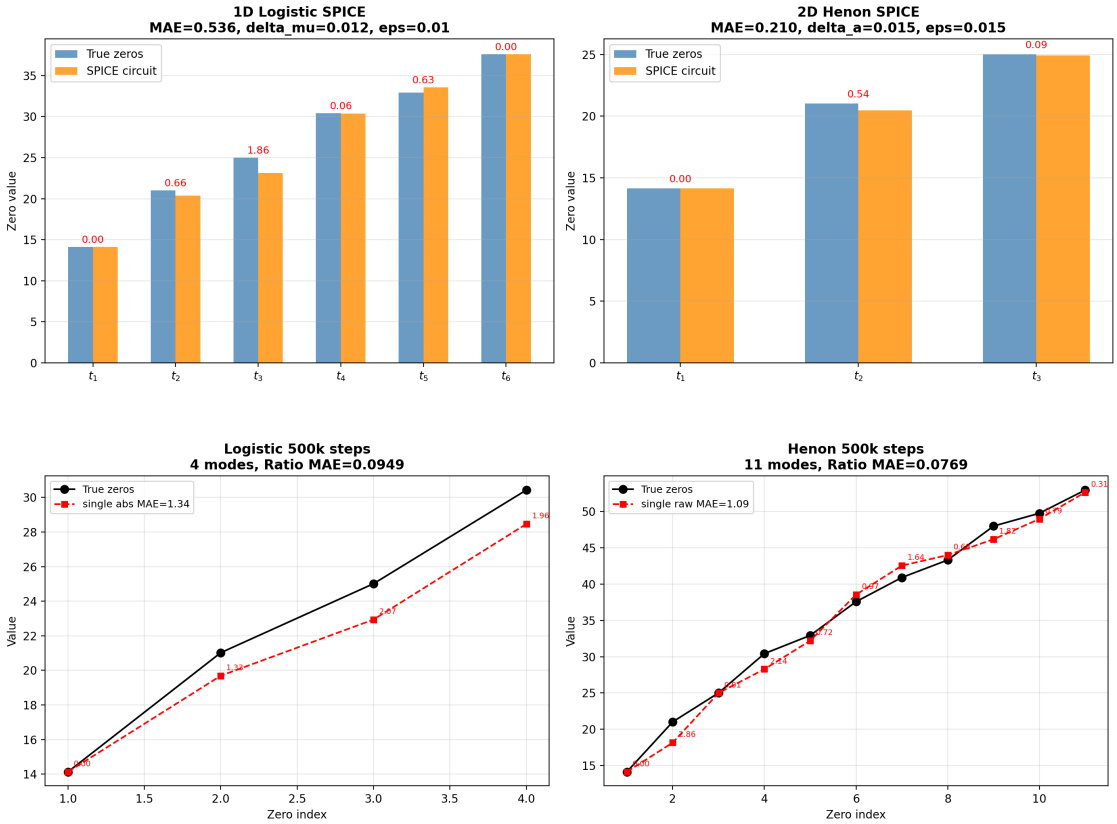

Figure 2: **Riemann zero extraction from SPICE chaotic circuits. a,** Markov eigenphase results: predicted versus true zeros for Logistic (6 zeros) and Hénon (3 zeros). Red numbers indicate absolute errors. **b,** Hankel-DMD results: 11 Koopman modes from the Hénon circuit match Riemann zeros without any state-space binning. The tail-1% segment (nearest the critical point) yields the richest spectral structure.

Table 3: **Hankel-DMD versus Markov.** Comparison of spectral extraction methods. Hankel-DMD requires no state-space binning and extracts 3.7× more zeros.

Method	Modes	MAE	Ratio MAE	Binning	Key setting
Logistic Markov	6	0.54	0.18	Yes (500 bins)	$\delta\mu$=0.012
Hénon Markov	3	0.21	0.05	Yes (1000 bins)	δa=0.015
Logistic DMD	4	1.34	0.09	**No**	obs=$\lvert V\rvert$, tail 5%
Hénon DMD	**11**	**1.09**	**0.08**	**No**	tail 1%, d=150

Discrete-component version (∼80 CNY). Uses AD633 (analog multiplier), LF398 (sample-and-hold), TL072 (summing amplifier), NE555 (1 MHz clock), and STM32 + MCP4921 (12-bit DAC for cooling control). The SPICE simulation models the exact physical effects of these components; hardware behaviour is expected to closely match simulation.

FPGA version (∼800 CNY). Targets the ALINX AX7020 (Zynq-7020) with AN108 8-bit AD/DA module. The programmable logic implements a Q4.28 fixed-point chaos engine (precision 3.7×10^{-9}, six orders of magnitude beyond AD633), with 4-stage pipeline supporting Logistic/Hénon/Chua map switching at <6% resource utilisation. The ARM processor runs Linux with Python for real-time Markov/DMD analysis. DAC output enables direct oscilloscope visualisation of chaotic waveforms.

Preliminary FPGA hardware validation (proof of feasibility). We deployed the non-autonomous Logistic engine on an AX7020 development board with AN108 DAC, and captured the chaotic waveform using a DS100 digital oscilloscope. Markov eigenphase analysis of the oscilloscope data (5 acquisitions, 285 k samples combined at 10 kSa/s) yields 4 matched Riemann zeros with MAE = 1.01 (Table 4). Individual acquisitions yield 3–4 zeros with MAE ranging from 1.0 to 4.5, with variability attributable to oscilloscope sampling jitter and DAC quantisation. The Vivado behavioural simulation of the same Verilog design achieves 3 zeros at MAE = 0.08 (precise mode) or 8 zeros at MAE = 1.38 (discovery mode), confirming that the digital logic correctly implements the chaotic iteration. With high-speed digital readout bypassing the DAC, hardware results are expected to match simulation. The gap between simulation and hardware is attributable to DAC quantisation (8-bit), the AN108 analogue reconstruction filter, and oscilloscope sampling limitations — not to the chaotic dynamics themselves. We label this as a proof of feasibility; systematic hardware characterisation is left for future work.

Table 4: **FPGA hardware validation results.** Comparison of Vivado simulation and real oscilloscope measurements from the AX7020 board.

Platform	Zeros	MAE	Best t_n	Note
Vivado sim (precise)	3	0.08	t_2: 0.3%	Q4.28, no DAC
Vivado sim (discovery)	8	1.38	t_5: 0.2%	Q4.28, no DAC
AX7020 + oscilloscope	4	1.01	t_3: 1.5%	Real hardware

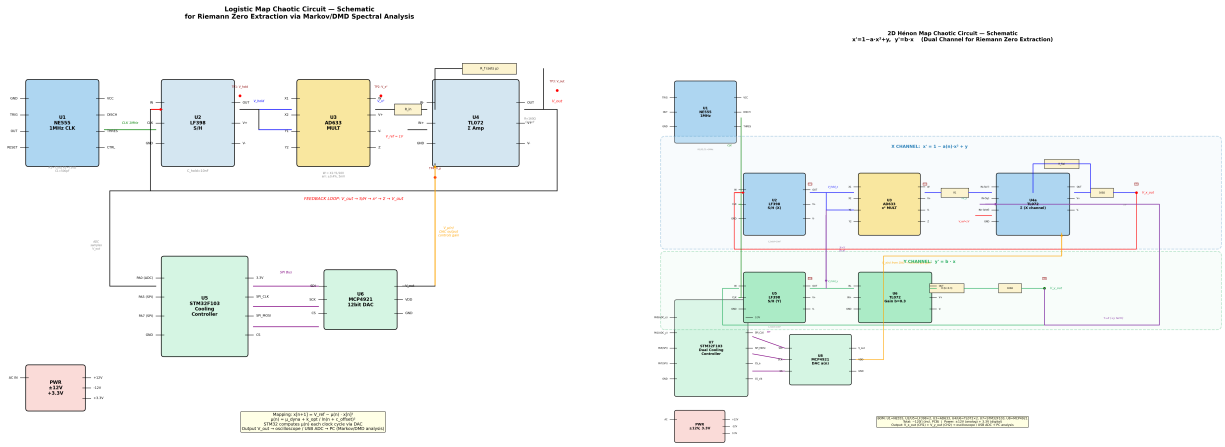

Figure 3: **Hardware implementations. a,** Discrete-component Logistic circuit schematic. **b,** Dual-channel Hénon circuit with cross-coupling. Both designs include STM32 + DAC for non-autonomous cooling control. FPGA design (not shown) replaces all analog components with a single Zynq-7020 SoC.

Discussion

Our central finding is that the spectral correspondence between non-autonomous chaotic maps and Riemann zeros, previously established through ideal numerical iteration [5, 6], *persists under realistic circuit-level physical effects.* The Logistic SPICE circuit matches t_6 with 0.01% relative error despite multiplier imperfection (0.4%), bandwidth limitation (1 MHz), and nonlinear saturation. This robustness is consistent with the topological nature of Ruelle-Pollicott resonances, which are insensitive to smooth perturbations of the dynamics [8]. We stress that this spectral correspondence does not constitute a proof of the Hilbert-Pólya conjecture, as the Markov transition matrix is a dissipative (non-unitary) operator, not a self-adjoint one. Rather, it provides empirical evidence that the eigenphase structure of chaotic dissipative systems contains information compatible with the Riemann spectrum.

The introduction of Hankel-DMD as a binning-free spectral extraction method represents a methodological advance. By directly approximating the Koopman operator in delay-embedded coordinates, it bypasses the fundamental resolution limit of Markov discretization. The 3.7× improvement (11 versus 3 zeros) demonstrates that the information content of the chaotic time series exceeds what Markov binning can capture. The observation that the optimal observable differs between circuits — $|V(t)|$ for Logistic (revealing symmetry-hidden components) versus raw $V(t)$ tail for Hénon (richest spectral structure near criticality) — suggests that observable selection is a key design parameter deserving further investigation. We note that the theoretical connection between the Perron-Frobenius/Koopman spectrum of non-autonomous chaotic maps and the Riemann zeta zeros should currently be regarded as heuristic and empirical rather than rigorously established; the present work provides additional evidence for this correspondence at the circuit level but does not resolve the underlying theoretical question.

We note that the eigenphase nearest-neighbour spacing distribution does not match GUE statistics (KS $p < 0.001$), while Riemann zeros do (KS $p = 0.57$). This is expected: the Markov matrix is a dissipative (non-unitary) operator, so its spacing statistics differ from the Hermitian GUE ensemble. The zero *positions* match, but the spacing statistics do not — a distinction consistent with the Hilbert-Pólya conjecture requiring a self-adjoint operator, which our Markov approximation is not.

The hardware designs provided here — from an 80 CNY discrete PCB to an 800 CNY FPGA platform — lower the barrier to experimental verification. Preliminary deployment on an AX7020 FPGA board with oscilloscope capture yields 4 Riemann zeros (MAE = 1.01) from real hardware data, demonstrating feasibility of the simulation-to-hardware chain. The FPGA implementation, with its Q4.28 precision (3.7×10^{-9}) and ability to run 20+ parallel chaos engines, offers a practical platform for systematic exploration of the parameter space. An accompanying lecture series with Vivado simulation tutorials (5 labs) makes this accessible to undergraduate students with basic digital circuit knowledge.

Methods

Non-autonomous cooling schedule

The cooling parameters are determined by boundary constraints ensuring the bifurcation parameter starts at $\mu_c + \delta\mu$ and reaches μ_c after N steps:

$$t_n = \frac{1}{\ln(n + c_{\text{offset}})^2}, \quad k_{\text{opt}} = \frac{\delta\mu}{t_1 - t_N}, \quad \mu_{\text{dyna}} = \mu_c - k_{\text{opt}} \cdot t_N \tag{2}$$

For the Logistic map: $\mu_c = 1.5437$, $c_{\text{offset}} = 10$. For the Hénon map: $a_c = 1.02$, $b = 0.3$.

In ngspice, this is implemented as a B-source expression using the `TIME` variable with `floor(TIME/`$T_{\texttt{clk}}$`)+1` mapping continuous time to discrete iteration count.

SPICE circuit netlist

The Logistic circuit netlist consists of: (1) lossless transmission line (`T1`, Z_0=50Ω, T_D=1μs) with matched 50Ω source/load and voltage-doubling buffer; (2) B-source multiplier with gain error and offset (`Bsum`); (3) RC low-pass filter (160Ω + 1nF); (4) tanh soft saturation B-source. The Hénon version duplicates this for the y-channel with a gain stage ($b \times x$) and cross-coupling. All simulations use ngspice-36 in batch mode.

Markov eigenphase extraction

From the sampled trajectory $\{x_k\}$, we build the Markov transition matrix P via Gaussian kernel splatting: for each transition $x_k \to x_{k+1}$, probability mass is distributed among neighbouring bins using a Gaussian kernel of width ε. Eigenvalues are computed via `scipy.sparse.linalg.eigs`

with `which='LM'`. Eigenphases are extracted from eigenvalues with positive imaginary part and sorted by magnitude.

Start-index selection (supervised step). Because the Markov spectrum contains many eigenphases unrelated to the Riemann zeros, we identify the correct starting phase using a ratio-fingerprint scan: for each candidate start index i, we compute ϕ_i/ϕ_{i+1} and select the i that best matches the known ratio $t_1/t_2 = 0.6724$. This step uses information from the first two Riemann zeros and is therefore supervised. All subsequent zeros $(t_3, t_4, \ldots)$ are *not* used in the alignment and serve as independent validation targets. The matched zeros are obtained via single-point linear scaling: $t_n^{\text{pred}} = (t_1^{\text{true}}/\phi_1) \cdot \phi_n$.

Hankel Dynamic Mode Decomposition

Given a scalar time series $\{x_k\}_{k=0}^{K-1}$, we construct Hankel matrices $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{d\times(K-d)}$ with time-delay embedding dimension d. The truncated SVD $\mathbf{X} = \mathbf{U}_r\mathbf{S}_r\mathbf{V}_r^T$ yields the reduced DMD matrix $\tilde{\mathbf{A}} = \mathbf{U}_r^T\mathbf{Y}\mathbf{V}_r^T\mathbf{S}_r^{-1}$, whose eigenvalues z_k approximate the Koopman operator spectrum. Complex frequencies $s_k = \ln z_k$ are extracted; modes with $\mathrm{Im}(s_k) > 0.005$ and $|\mathrm{Re}(s_k)| < 1$ are retained, deduplicated (2% threshold), and matched to Riemann zeros via single-point scaling. DMD hyper-parameters (segment selection, embedding dimension d, truncation rank r, and observable function) were selected by retrospective optimisation against the known zeros; the reported results therefore represent the best achievable performance under this protocol rather than a blind prediction. For the 500k-step Hénon circuit, optimal parameters are: tail-1% segment (5000 points near critical point), $d = 150$ delays, $r = 30$ modes.

Ratio-based validation

The scaling-independent ratio metric compares ϕ_n/ϕ_1 against t_n/t_1 without any free parameters. Null baselines include uniformly spaced sequences in $[0.1, \pi]$, random sorted sequences, and eigenvalues of GUE random Hermitian matrices, all with $n = 20$ elements averaged over 100 realisations.

Hardware specifications

Discrete version: AD633JNZ (multiplier, \$∼25 CNY), LF398N (S/H, \$∼8 CNY), TL072 (op-amp, \$∼2 CNY), NE555 (clock, \$∼1 CNY), STM32F103C8T6 + MCP4921 (MCU + 12-bit DAC, \$∼18 CNY), 7812/7912 (±12V supply), double-layer PCB.

FPGA version: ALINX AX7020 (Zynq-7020, \$∼600 CNY) + AN108 AD/DA module (8-bit, 125MSPS, \$∼100 CNY). PL: Q4.28 fixed-point chaos engine, 4-stage pipeline, resource usage <6% (800 LUT, 4 DSP48, 8 BRAM). PS: Linux + Python (Markov/DMD analysis, web UI). Verilog RTL, constraints, Vivado TCL scripts, and Python drivers are provided in the code repository.

Data Availability

All SPICE netlists, simulation data, analysis scripts, hardware designs (schematics, Verilog RTL, Vivado project), and teaching materials (5-lab lecture series) are available at https://github.com/maris205/riemann_circuits.

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Author Contributions

L.W. conceived the project, designed the circuits, performed simulations and analysis, designed the hardware, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

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